

AIR UNIVERSITY, AEROSPACE AND AVIATION CAMPUS, KAMRA

NAME	**EIMAN FATIMA**
REG ID	**245237**
SUBMITTED TO	**MENAHIL NASEER**
LAB	**4**
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Task 1: Navigation and Routing: A Pizza Store App

You are building the navigation skeleton of a Pizza Store app. The app should let a customer browse the menu,

view the details of a specific pizza, and add it to their cart.

○ HomePage: a menu list of at least four pizzas, each built from Container, Row, Column, Image, and Icon

widgets (photo, name, price, a star-rating icon).

○ Tapping a pizza on HomePage navigates to a DetailsPage using Navigator.push() + MaterialPageRoute, passing

the selected pizza's name, price, and image URL through the constructor.

○ DetailsPage shows the full details and includes an ElevatedButton ('Add to Cart') that calls

Navigator.pop(context, itemName) to return the selected pizza's name to HomePage.

○ HomePage awaits the Navigator.push() call and shows a confirmation (e.g., a SnackBar reading 'Added

<name> to cart') using the returned value.

○ Add a FloatingActionButton on HomePage that opens a CartPage (a simple placeholder screen is fine) using a

named route ('/cart').

○ Wrap the pizza image in a Hero widget (matching tag) on both HomePage and DetailsPage so it animates

smoothly between the two screens.

Deliverable: working source code, plus a short screen recording or screenshots showing the full flow — Home →

Details → back to Home with a confirmation → Cart.

CODE:

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
      routes: {
        '/cart': (context) => const CartPage(),
      },
    );
  }
}
class HomePage extends StatelessWidget {
  const HomePage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Pizza Store'),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.pushNamed(context, '/cart');
        },
        child: const Icon(Icons.shopping_cart),
      ),
```

```
      body: ListView(
        children: [
          pizzaCard(
            context,
            'Margherita Pizza',
            'Rs. 800',
            'https://picsum.photos/id/292/300/300',
          ),
          pizzaCard(
            context,
            'Pepperoni Pizza',
            'Rs. 1000',
            'https://picsum.photos/id/326/300/300',
          ),
          pizzaCard(
            context,
            'Cheese Pizza',
            'Rs. 750',
            'https://picsum.photos/id/429/300/300',
          ),
          pizzaCard(
            context,
            'Chicken Pizza',
            'Rs. 1200',
            'https://picsum.photos/id/488/300/300',
          ),
        ],
      ),
    );
  }
```

```dart
Widget pizzaCard(
    BuildContext context,
    String name,
    String price,
    String image,
    ) {
  return GestureDetector(
    onTap: () async {
      var result = await Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => DetailsPage(
            name: name,
            price: price,
            image: image,
          ),
        ),
      );
      if (result != null) {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(
            content: Text('Added $result to cart'),
          ),
        );
      }
    },
    child: Container(
      height: 150,
      padding: const EdgeInsets.all(16),
      margin: const EdgeInsets.all(10),
```

```
92                  decoration: BoxDecoration(
93                    color: Colors.white,
94                    borderRadius: BorderRadius.circular(10),
95                    boxShadow: const [
96                      BoxShadow(
97                        color: Colors.grey,
98                        blurRadius: 5,
99                      ),
100                   ],
101                 ),
102                 child: Row(
103                   children: [
104                     Image.network(
105                       image,
106                       width: 100,
107                       height: 100,
108                       fit: BoxFit.cover,
109                     ),
110                     const SizedBox(width: 15),
111                     Column(
112                       mainAxisAlignment: MainAxisAlignment.center,
113                       crossAxisAlignment: CrossAxisAlignment.start,
114                       children: [
115                         Text(
116                           name,
117                           style: const TextStyle(
118                             fontSize: 18,
119                             fontWeight: FontWeight.bold,
120                           ),
121                         ),
122                         const SizedBox(height: 10),
```

```
                    Text(price),
                    const SizedBox(height: 10),
                    const Row(
                      children: [
                        Icon(
                          Icons.star,
                          color: Colors.amber,
                          size: 18,
                        ),
                        Text(' 4.8 rating'),
                      ],
                    ),
                  ],
                ),
              ],
            ),
          ),
        ),
      );
    }
  }
  class DetailsPage extends StatelessWidget {
    final String name;
    final String price;
    final String image;
    const DetailsPage({
      super.key,
      required this.name,
      required this.price,
      required this.image,
    });
    @override
```

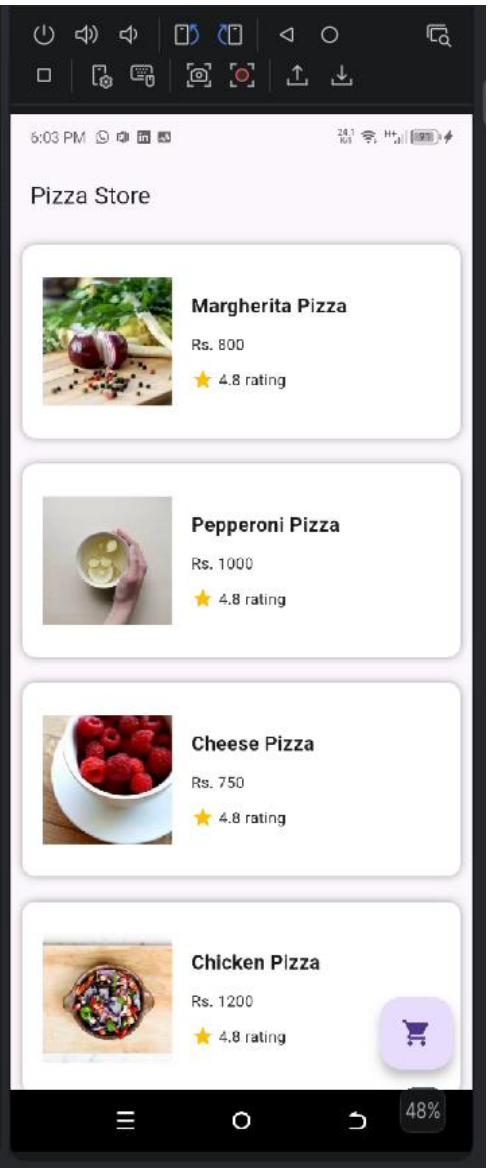
```dart
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Pizza Details'),
    ),
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Image.network(
            image,
            width: 250,
            height: 250,
            fit: BoxFit.cover,
          ),
          const SizedBox(height: 20),
          Text(
            name,
            style: const TextStyle(
              fontSize: 24,
              fontWeight: FontWeight.bold,
            ),
          ),
          const SizedBox(height: 10),
          Text(
            price,
            style: const TextStyle(
              fontSize: 20,
            ),
          ),
```

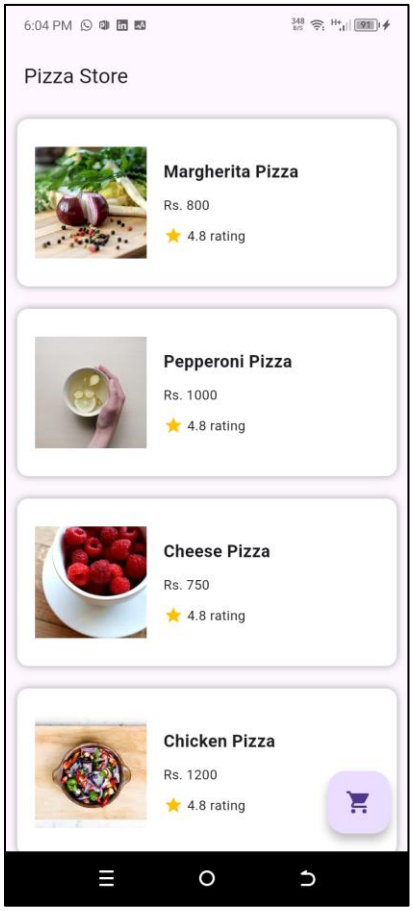
```
                const SizedBox(height: 20),
                ElevatedButton(
                  onPressed: () {
                    Navigator.pop(context, name);
                  },
                  child: const Text('Add to Cart'),
                ),
              ],
            ),
          ),
        );
      }
    }
    class CartPage extends StatelessWidget {
      const CartPage({super.key});
      @override
      Widget build(BuildContext context) {
        return Scaffold(
          appBar: AppBar(
            title: const Text('Cart'),
          ),
          body: const Center(
            child: Text(
              'Your cart is empty',
              style: TextStyle(
                fontSize: 20,
              ),
            ),
          ),
        );
```

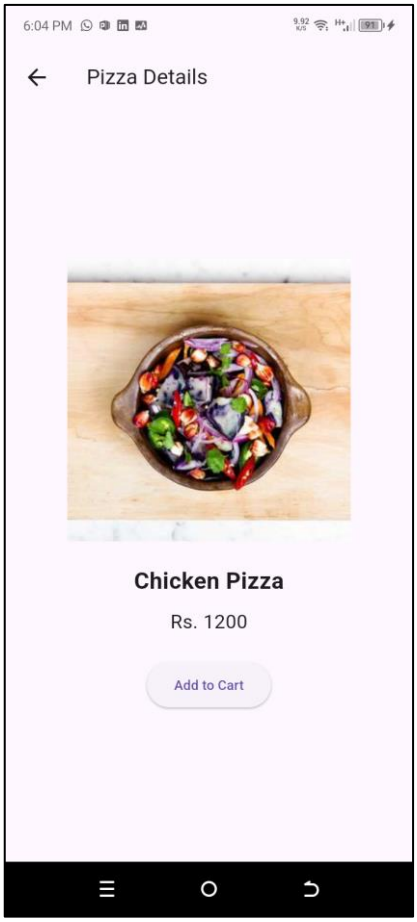
```
        );
      }
    }
```

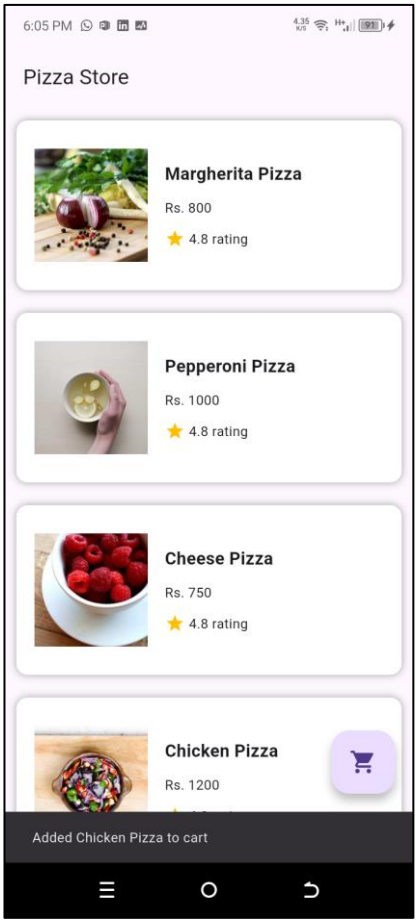
OUTPUT:

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
      routes: {
        '/cart': (context) => const CartPage(),
      },
    );
  }
}
class HomePage extends StatelessWidget {
  const HomePage({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Pizza Store'),
      ),
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          Navigator.pushNamed(context, '/cart');
        },
        child: const Icon(Icons.shopping_cart),
      ),
```









Task 2: Multi-Stage Quiz Application

You are developing a multi-stage quiz application using Flutter. The app is designed to help users test their knowledge across various topics.

○ Implement named routes for each stage of the quiz: HomePage, QuizPage, ReviewPage, and ResultsPage.

○ Use Navigator.pushNamed() to move forward between stages, and Navigator.pop() to let the user return to aprevious stage when needed (e.g., from ReviewPage back to QuizPage to change an answer).

○ Pass the user's selected answers forward between screens either through each page's constructor.

○ ResultsPage should compute and display a final score based on the answers received.Deliverable: working source code, plus a short explanation of how data flows from QuizPage through to ResultsPage.

CODE:

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      initialRoute: '/',
      routes: {
        '/': (context) => const HomePage(),
        '/quiz': (context) => const QuizPage(),
        '/review': (context) => const ReviewPage(),
        '/results': (context) => const ResultsPage(),
      },
    );
  }
}

class HomePage extends StatelessWidget {
```

```dart
const HomePage({super.key});

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: const Text('Quiz App'),
    ),
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          const Text(
            'Welcome to the Quiz',
            style: TextStyle(
              fontSize: 24,
              fontWeight: FontWeight.bold,
            ),
          ),
          const SizedBox(height: 20),
          ElevatedButton(
            onPressed: () {
              Navigator.pushNamed(context, '/quiz');
            },
            child: const Text('Start Quiz'),
          ),
        ],
      ),
    ),
  );
}
```

```dart
}

class QuizPage extends StatefulWidget {
  const QuizPage({super.key});

  @override
  State<QuizPage> createState() => _QuizPageState();
}

class _QuizPageState extends State<QuizPage> {
  String? a1;
  String? a2;
  String? a3;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Quiz'),
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          crossAxisAlignment: CrossAxisAlignment.start,
          children: [
            const Text('Q1: What is the capital of Pakistan?'),
            DropdownButton<String>(
              hint: const Text('Select answer'),
              value: a1,
              items: ['Islamabad', 'Lahore', 'Karachi']
                  .map(
```

```
            (s) => DropdownMenuItem(
              value: s,
              child: Text(s),
            ),
          )
              .toList(),
          onChanged: (val) {
            setState(() {
              a1 = val;
            });
          },
        ),
        const SizedBox(height: 20),
        const Text('Q2: Which language is used in Flutter?'),
        DropdownButton<String>(
          hint: const Text('Select answer'),
          value: a2,
          items: ['Dart', 'Java', 'Kotlin']
              .map(
                (s) => DropdownMenuItem(
              value: s,
              child: Text(s),
            ),
          )
              .toList(),
          onChanged: (val) {
            setState(() {
              a2 = val;
            });
          },
        ),
```

```
const SizedBox(height: 20),
const Text('Q3: Which widget stacks children vertically?'),
DropdownButton<String>(
  hint: const Text('Select answer'),
  value: a3,
  items: ['Column', 'Row', 'Stack']
      .map(
        (s) => DropdownMenuItem(
      value: s,
      child: Text(s),
    ),
  )
      .toList(),
  onChanged: (val) {
    setState(() {
      a3 = val;
    });
  },
),
const SizedBox(height: 30),
Center(
  child: ElevatedButton(
    onPressed: () {
      List<String> answers = [
        a1 ?? 'Not answered',
        a2 ?? 'Not answered',
        a3 ?? 'Not answered',
      ];
      Navigator.pushNamed(
        context,
        '/review',
```

```
                        arguments: answers,
                      );
                    },
                    child: const Text('Review Answers'),
                  ),
                ),
              ],
            ),
          ),
        );
      }
    }

    class ReviewPage extends StatelessWidget {
      const ReviewPage({super.key});

      @override
      Widget build(BuildContext context) {
        List<String> answers =
        ModalRoute.of(context)!.settings.arguments as List<String>;

        return Scaffold(
          appBar: AppBar(
            title: const Text('Review'),
          ),
          body: Padding(
            padding: const EdgeInsets.all(16),
            child: Column(
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text('Q1: ${answers[0]}'),
```

```
                const SizedBox(height: 10),
                Text('Q2: ${answers[1]}'),
                const SizedBox(height: 10),
                Text('Q3: ${answers[2]}'),
                const SizedBox(height: 30),
                Row(
                  mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                  children: [
                    OutlinedButton(
                      onPressed: () {
                        Navigator.pop(context);
                      },
                      child: const Text('Change Answers'),
                    ),
                    ElevatedButton(
                      onPressed: () {
                        Navigator.pushNamed(
                          context,
                          '/results',
                          arguments: answers,
                        );
                      },
                      child: const Text('Submit'),
                    ),
                  ],
                ),
              ],
            ),
          ),
        );
    }
```

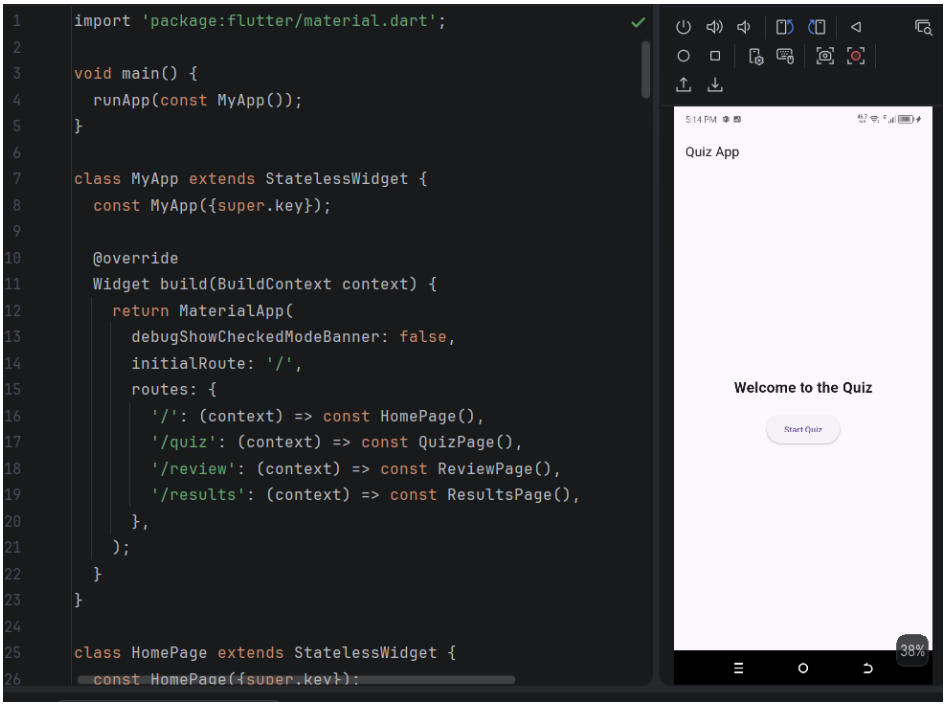
```dart
}
class ResultsPage extends StatelessWidget {
  const ResultsPage({super.key});
  @override
  Widget build(BuildContext context) {
    List<String> answers =
    ModalRoute.of(context)!.settings.arguments as List<String>;

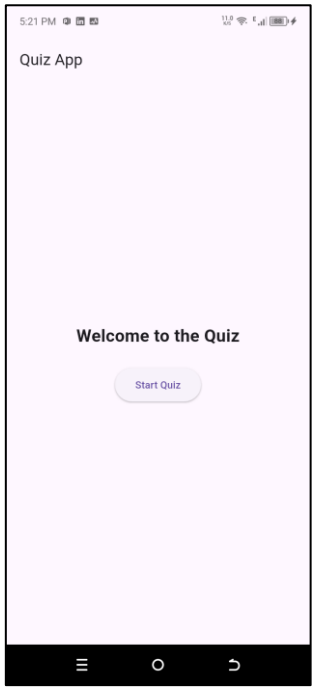
    int score = 0;

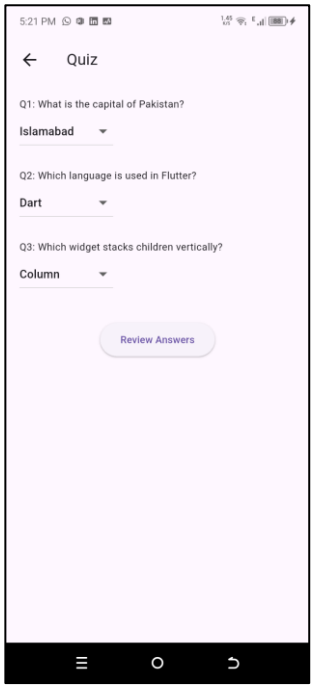
    if (answers[0] == 'Islamabad') {
      score++;
    }
    if (answers[1] == 'Dart') {
      score++;
    }
    if (answers[2] == 'Column') {
      score++;
    }
    return Scaffold(
      appBar: AppBar(
        title: const Text('Results'),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            const Text(
              'Your Score',
              style: TextStyle(
                fontSize: 24,
```

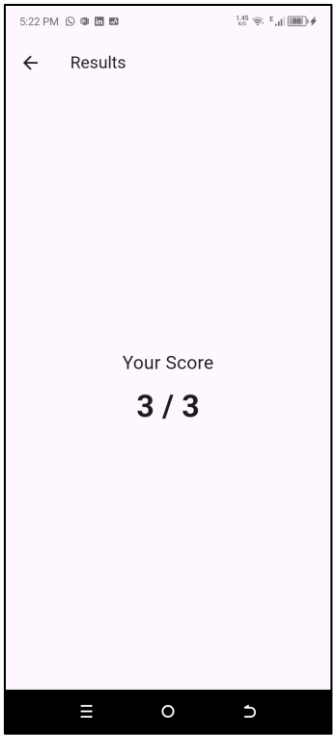
```
                ),
              ),
              const SizedBox(height: 10),
              Text(
                '$score / 3',
                style: const TextStyle(
                  fontSize: 40,
                  fontWeight: FontWeight.bold,
                ),
              ),
            ],
          ),
        ),
      );
    }
}
```

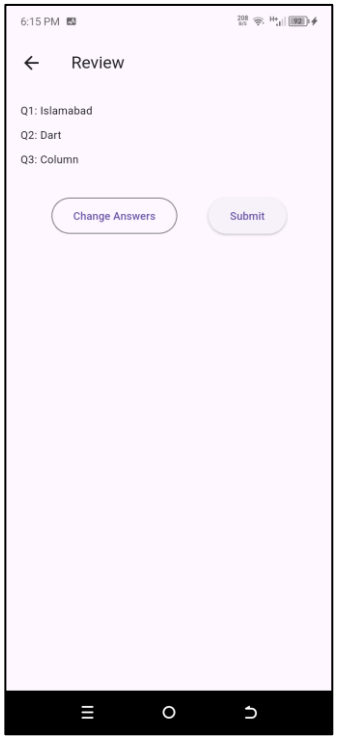
OUTPUT:




5:21 PM
Quiz App
Welcome to the Quiz
Start Quiz


5:21 PM
Quiz
Q1: What is the capital of Pakistan?
Islamabad
Q2: Which language is used in Flutter?
Dart
Q3: Which widget stacks children vertically?
Column
Review Answers


5:22 PM
Results
Your Score
3 / 3


6:15 PM
Review
Q1: Islamabad
Q2: Dart
Q3: Column
Change Answers
Submit

Data-flow explanation:

On QuizPage, each dropdown selection is stored in a variable (a1, a2, a3) using setState. When "Review Answers" is pressed, the three answers are put into a list and sent to ReviewPage using Navigator.pushNamed with the arguments parameter. ReviewPage reads the list using ModalRoute.of(context)!.settings.arguments and displays it. "Change Answers" calls Navigator.pop() to go back to QuizPage. "Submit" sends the same list to ResultsPage the same way. ResultsPage compares each answer with the correct one, calculates the score and displays it.

Task 3: Pet Adoption Explorer

Develop a multi-screen Pet Adoption Explorer app:

o The home screen should display at least four pets using Container, Row, Column, Image, and Icon widgets. Each pet should display its image, name, type, and rating.

o When a pet is selected, navigate to a Pet Details screen using Navigator.push() and MaterialPageRoute. Pass

the selected pet's name, image, and other basic information to the Details screen.

o The Details screen should contain an Adopt Me button. When the button is pressed, use Navigator.pop() to return an adoption message to the Home screen. Display the returned message using a SnackBar.

o Add a Hero animation to the pet image so that the image smoothly moves between the Home and Details screens.

CODE:

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Pet Adoption Explorer'),
      ),
      body: ListView(
        children: [
          petCard(
            context,
```

```
            'Bella',
            'Dog',
            '4.9',
            'https://picsum.photos/id/237/300/300',
          ),
          petCard(
            context,
            'Milo',
            'Cat',
            '4.7',
            'https://picsum.photos/id/40/300/300',
          ),
          petCard(
            context,
            'Max',
            'Dog',
            '4.8',
            'https://picsum.photos/id/1025/300/300',
          ),
          petCard(
            context,
            'Luna',
            'Cat',
            '4.6',
            'https://picsum.photos/id/1074/300/300',
          ),
        ],
      ),
    );
  }
```

```dart
Widget petCard(
    BuildContext context,
    String name,
    String type,
    String rating,
    String image,
    ) {
  return GestureDetector(
    onTap: () async {
      var result = await Navigator.push(
        context,
        MaterialPageRoute(
          builder: (context) => PetDetailsPage(
            name: name,
            type: type,
            rating: rating,
            image: image,
          ),
        ),
      );

      if (result != null) {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(
            content: Text(result),
          ),
        );
      }
    },
    child: Container(
      height: 150,
```

```
        padding: const EdgeInsets.all(16),
        margin: const EdgeInsets.all(10),
        decoration: BoxDecoration(
          color: Colors.white,
          borderRadius: BorderRadius.circular(10),
          boxShadow: const [
            BoxShadow(
              color: Colors.grey,
              blurRadius: 5,
            ),
          ],
        ),
        child: Row(
          children: [
            Image.network(
              image,
              width: 100,
              height: 100,
              fit: BoxFit.cover,
            ),
            const SizedBox(width: 15),
            Column(
              mainAxisAlignment: MainAxisAlignment.center,
              crossAxisAlignment: CrossAxisAlignment.start,
              children: [
                Text(
                  name,
                  style: const TextStyle(
                    fontSize: 18,
                    fontWeight: FontWeight.bold,
                  ),
```

```
                ),
                const SizedBox(height: 10),
                Text(type),
                const SizedBox(height: 10),
                Row(
                  children: [
                    const Icon(
                      Icons.star,
                      color: Colors.amber,
                      size: 18,
                    ),
                    Text(' $rating rating'),
                  ],
                ),
              ],
            ),
          ],
        ),
      ),
    );
  }
}

class PetDetailsPage extends StatelessWidget {
  final String name;
  final String type;
  final String rating;
  final String image;

  const PetDetailsPage({
```

```
155       super.key,
156       required this.name,
157       required this.type,
158       required this.rating,
159       required this.image,
160     });
161
162     @override
163     Widget build(BuildContext context) {
164       return Scaffold(
165         appBar: AppBar(
166           title: const Text('Pet Details'),
167         ),
168         body: Center(
169           child: Column(
170             mainAxisAlignment: MainAxisAlignment.center,
171             children: [
172               Image.network(
173                 image,
174                 width: 250,
175                 height: 250,
176                 fit: BoxFit.cover,
177               ),
178               const SizedBox(height: 20),
179               Text(
180                 name,
181                 style: const TextStyle(
182                   fontSize: 24,
183                   fontWeight: FontWeight.bold,
184                 ),
185               ),
```

```
            const SizedBox(height: 10),
            Text(
              type,
              style: const TextStyle(
                fontSize: 20,
              ),
            ),
            const SizedBox(height: 10),
            Row(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                const Icon(
                  Icons.star,
                  color: Colors.amber,
                  size: 20,
                ),
                Text(' $rating rating'),
              ],
            ),
            const SizedBox(height: 20),
            ElevatedButton(
              onPressed: () {
                Navigator.pop(context, 'You have adopted $name!');
              },
              child: const Text('Adopt Me'),
            ),
          ],
        ),
      ),
    );
  }
```

```
}
```

OUTPUT:

```dart
import 'package:flutter/material.da
1

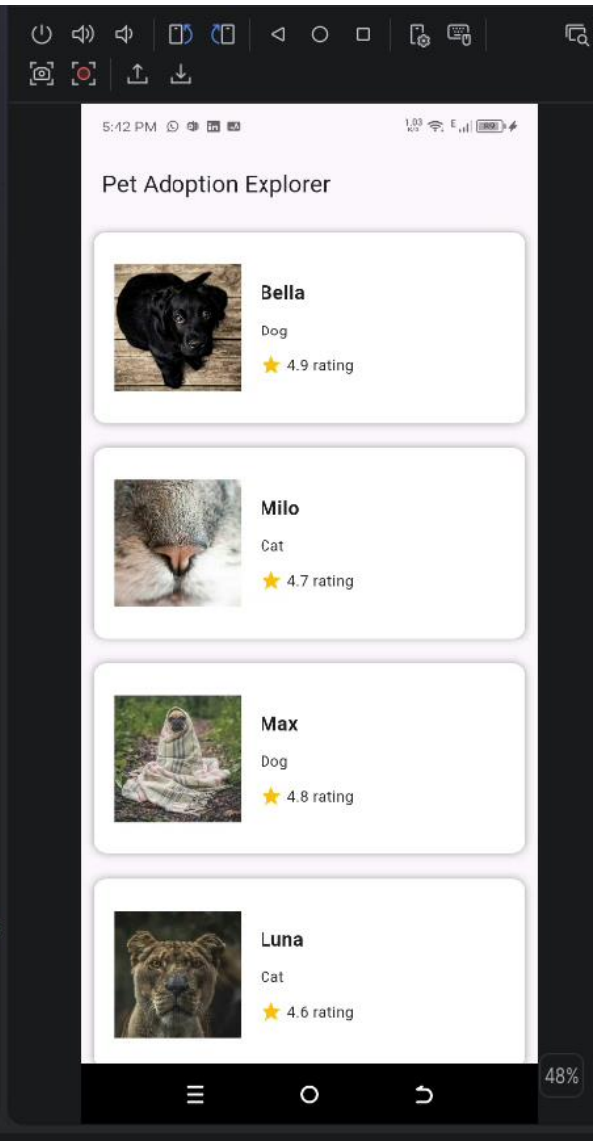
void main() {
  runApp(const MyApp());
}

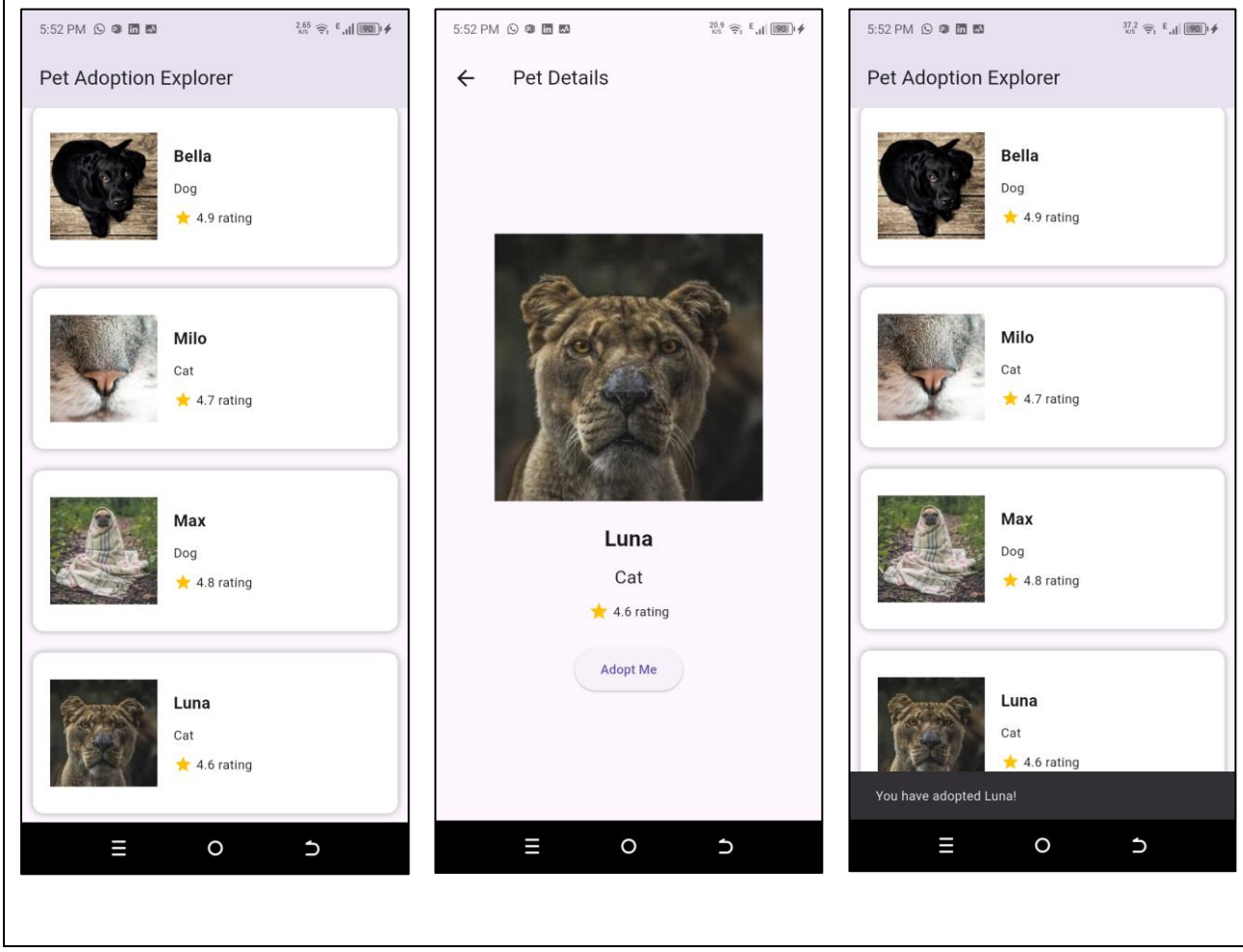
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Pet Adoption Explo
      ),
      body: ListView(
        children: [
          petCard(
            context,
```




5:52 PM
Pet Adoption Explorer
Bella
Dog
4.9 rating
Milo
Cat
4.7 rating
Max
Dog
4.8 rating
Luna
Cat
4.6 rating
5:52 PM
Pet Details
Luna
Cat
4.6 rating
Adopt Me
5:52 PM
Pet Adoption Explorer
Bella
Dog
4.9 rating
Milo
Cat
4.7 rating
Max
Dog
4.8 rating
Luna
Cat
4.6 rating
You have adopted Luna!